IEEJ TRANSACTIONS ON ELECTRICAL AND ELECTRONIC ENGINEERING
IEEJ Trans 2013; **8**: 640–648
Published online in Wiley Online Library (wileyonlinelibrary.com). DOI:10.1002/tee.21908

Paper

Improvement of Saturation Property of Iron Powder Core by Flux Homogenizing Structure

Kazuhiro Umetani[a], Member

A new core structure is proposed to improve the saturation property of an inductor made of iron powder core. In the conventional structure made of a single core material, converging flux lines cause local magnetic saturation at the inner side of the corners of the magnetic path. Therefore, the conventional inductor shows worse saturation property than expected from its core material. This degradation mechanism is solved in the proposed core structure through homogenization of the flux distribution at the corners. Experiments are performed to evaluate the flux homogenization and improvement of the saturation property. Results show a homogenized flux distribution in the proposed core structure. Furthermore, the flux level improved by 28% while the inductance decreased by 40% of its initial value. Both these results are consistent with theoretical expectations, except that a larger improvement is found in the experiment compared to the 16% improvement predicted by numerical calculation. © 2013 Institute of Electrical Engineers of Japan. Published by John Wiley & Sons, Inc.

Keywords: iron powder core, saturation property, flux distribution

Received 19 June 2012; revised 8 October 2012

1. Introduction

In the design of power conversion circuits, improving the trade-off between higher efficiency and less material consumption is constantly attempted. An inductor, one of the major voluminous components, is also studied in order to achieve sufficient magnetic property by using the minimum amount of material. Since an inductor stores its energy in its magnetic path, the magnetic property of the inductor is closely related to the core. Thus, the major focus is on improving the inductor core.

An inductor core has a closed magnetic path made of soft magnetic material and gaps. Because of its simple structure, there is not much scope for its improvement. While a number of studies [1–3] have attempted to develop materials with preferred characteristics, investigation of a better core structure is another possible approach.

This approach is probably effective, especially when the magnetic material has low permeability, such as an iron powder core [4]. An inductor with an iron powder core generally does not have gaps because of its low permeability. Accordingly, its inductance directly reflects the magnetic resistance of the core material itself, which is severely affected by the local distribution of magnetic flux density in the core.

Because the inner side of a magnetic path is inevitably shorter than the outer side, magnetic flux in the core is generally distributed inhomogeneously. The flux density tends to be large at the inner side and small at the outer side. Especially, pronounced inhomogeneity has been reported at a corner of a rectangular magnetic path [5].

Converging flux at the inner side promotes magnetic saturation, which increases the magnetic resistance of the inner path. In addition, sparse distribution at the outer side leads to less efficient energy storage in the outer path.

Consequently, the inhomogeneous distribution of flux density worsens the saturation property of the inductor and reduces the maximum electric current. This results in the reduction of the maximum energy storable in the core. Since the energy stored is approximately proportional to the volume of the core, upsizing the core is required in order to compensate for the degradation of the property.

With a view to solving the problem, this paper proposes a novel core structure that homogenizes the radial distribution of flux density. In other words, the inner path has the same flux density as the outer path in the proposed core. Since the structure resolves flux inhomogeneity and the resulting degradation mechanism, it is expected to improve the saturation property of an inductor. The improvement will result in the reduction of core material because the necessary saturation property is achievable without compensatory upsizing.

Along with the proposal, this paper also presents both numerical and experimental confirmations of both homogeneous distribution and improvement of the saturation property. The next section discusses theoretical derivation of the proposed core structure. Numerical calculations of the magnetic property are presented in Section 3, and experimental results are given in Section 4.

2. Proposed Core Structure

2.1. Derivation of core structure This section derives the physical core structure in which radial flux distribution is homogenized. To simplify our discussion, we do not consider magnetic saturation, leakage flux, or electric and displacement currents inside magnetic materials. We assume that the core structure is uniform perpendicular to the flux path. Therefore, we discuss two-dimensional (2-D) flux vectors. In addition, we assume that each material constituting the proposed core has constant permeability.

As a possible approach to obtain homogeneous flux distribution in the radial direction, we consider a closed magnetic path of series-connected pieces of core materials in which the flux density is distributed homogeneously. Then, the basic outline of the proposed core structure is derived by considering the required shape of flux lines in each piece of core material.

[a] Correspondence to: Kazuhiro Umetani.
E-mail: KAZUHIRO_Z_UMETANI@denso.co.jp

DENSO CORPORATION Research Laboratories, 500-1 Minamiyama, Komenoki-cho, Nisshin City, Aich Pref. 470-0111, Japan

© 2013 Institute of Electrical Engineers of Japan. Published by John Wiley & Sons, Inc.

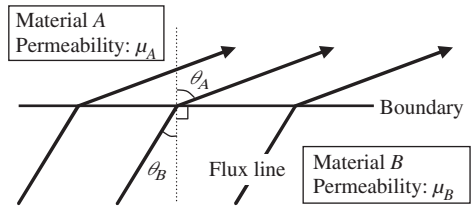


Fig. 1. Flux refraction at the boundary of magnetic materials

According to static magnetism [6], the magnetic flux vector in a piece of core material is described by the following equations:

$$\nabla \cdot \mathbf{B} = 0 \tag{1}$$

$$\nabla \times \left(\frac{\mathbf{B}}{\mu} \right) = 0 \tag{2}$$

where $\mathbf{B}$ is the magnetic flux density, and μ is the permeability.

The above equations lead to the fact that a 2-D homogeneous field with constant permeability should form straight flux lines. The details are shown in Appendix A1. Since homogeneous flux distribution forms straight flux lines in each piece of core material, the flux lines in the proposed core are required to form polygons. Consequently, the magnetic path of the proposed core structure is not allowed to be toroidal; instead, it should be polygonal.

The question then arises about the structure of a corner path. Because an angle in a flux line inevitably destroys homogeneity anywhere inside the medium, the boundary is instead utilized to make the angle.

According to magnetism [7], a flux line that passes through materials of different permeability, as illustrated in Fig. 1, is known to turn at the boundary surface in a manner similar to the refraction of light. The *incident* and *refraction* angles θ_A and θ_B are constrained by the following relation:

$$\frac{\tan\theta_A}{\tan\theta_B} = \frac{\mu_A}{\mu_B} \tag{3}$$

Consequently, the turning angle is obtained as $\theta_A - \theta_B$. An angle smaller than 90° is made by a single refraction process, since $0^\circ < \theta_A, \theta_B < 90^\circ$.

To utilize the phenomenon in the physical core, the structure shown in Fig. 2 is employed. The corner structure, as illustrated by the thick line, is composed of two right triangles that have a common hypotenuse. The triangular pieces are made of materials with different permeability μ_A and μ_B so that the vertex angles θ_A and θ_B suffice (3). According to the refraction rule, the *incident* flux line perpendicular to side B is bent perpendicular to side A. In brief, the corner structure forms an angle on the flux line.

Equation (3) also shows the required shape of the magnetic path that extends outside sides A and B. Because the flux lines

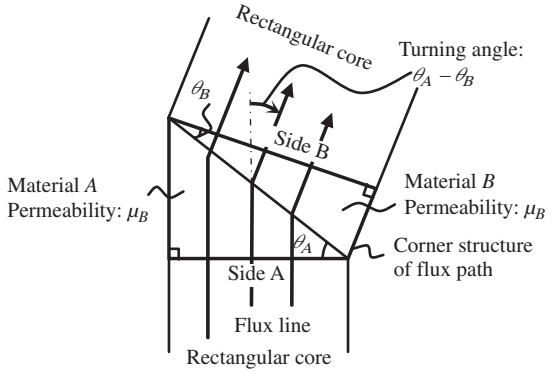


Fig. 2. Corner of the proposed core structure

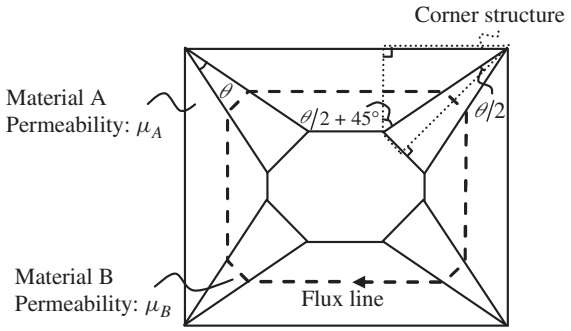


Fig. 3. Example of the proposed core with rectangular outline

perpendicular to side A or B do not refract regardless of the permeability μ_A or μ_B, the extended paths should let them pass. Therefore, a piece of core that contacts the corner structure should be a rectangle or another corner structure.

By composing a closed path from the rectangular pieces and the corner structures, a polygonal path made of straight flux lines is finally obtained. We are able to obtain magnetic paths of various polygons by designing appropriate turning angles of the corner structures.

As an example, an inductor core with a rectangular outline is presented in Fig. 3. The magnetic path forms a right angle at the corners. Since the angle cannot be obtained by a single corner structure, the corner is made of two corner structures connected in series. The angle θ, along with the permeability μ_A and μ_B, is determined after the following equation, which is derived from (3):

$$\frac{\tan(45^\circ + \theta/2)}{\tan(\theta/2)} = \frac{\mu_A}{\mu_B} \tag{4}$$

The same material is assigned for pieces that contact the boundary perpendicular to the flux lines. As a result, two materials suffice to constitute the core.

2.2. Theoretical confirmation of homogeneity In the previous section, the required structure of the physical core is derived from the assumption that homogeneous flux distribution is given. Thus, the result is only a necessary condition for homogeneous distribution. This section shows its sufficiency by discussing that the assumption is certainly valid in the proposed structure.

Suppose that the flux line is polygonal, as described in the previous subsection, and that the norm of the flux density vector $\mathbf{B}$ is constant anywhere inside any piece of the core material. For convenience, the leakage flux is ignored, and there the permeability is approximated as zero. Then, this flux distribution is indeed the solution for the static magnetic system for the following reasons:

First, the supposed distribution satisfies (1) and (2) inside each piece of core material, because the permeability is constant and the space derivative of $\mathbf{B}$ vanishes.

Next, the boundary condition of the static magnetic system is satisfied at any boundary between the pieces. Taking the boundary shown in Fig. 1 as an example, the boundary conditions are described by the following equations [7,8]:

$$\mathbf{n} \cdot \mathbf{B}_A = \mathbf{n} \cdot \mathbf{B}_B \tag{5}$$

$$\mathbf{n} \times \left(\frac{\mathbf{B}_A}{\mu_A} \right) = \mathbf{n} \times \left(\frac{\mathbf{B}_B}{\mu_B} \right) \tag{6}$$

where $\mathbf{n}$ is the normal vector of the boundary surface, and $\mathbf{B}_A$ and $\mathbf{B}_B$ are the flux density in materials A and B, respectively.

Now, the supposed distribution satisfies the refraction rule, i.e. (3), at the boundary. Furthermore, the norm of $\mathbf{B}$ has the

19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

following relation, because the width of the magnetic path is proportional to the cosine of the refraction angle θ_A or θ_B:

$$|\mathbf{B}_A| \cos\theta_A = |\mathbf{B}_B| \cos\theta_B \tag{7}$$

The boundary condition is then satisfied because (3) and (7) are sufficient to satisfy (5) and (6).

As for the lateral boundary between the core and the atmosphere, **B** is parallel to the boundary. Therefore, (5) is satisfied automatically. Equation (6) is also satisfied if the atmospheric flux density B_{atm} is set at $\mu_{\text{atm}} \cdot B/\mu$, where μ_{atm} and μ are the permeability of the atmosphere and the core, respectively, and B is the flux density in the core. The atmospheric flux density B_{atm} vanishes finally by taking the limit of $\mu_{\text{atm}} \to 0$.

As discussed above, (1), (2), and the boundary conditions are found to be satisfied everywhere. Therefore, homogeneous distribution is the solution of the static magnetic system for the proposed core structure.

3. Numerical Calculations

3.1. Flux distribution To confirm the flux homogeneity, 2-D calculation of the flux distribution is performed. The proposed core, with the rectangular outline shown in Fig. 3, is compared to a conventional core with the same outline but consisting of a single material. Linear media with constant permeability are employed for all materials in the model. The permeability is determined so that the total flux induced in one core equals that of the other under the same magnetomotive force. In other words, we set the same AL value for the two cores. The simulator is Maxwell SV 9.0.

Figure 4 illustrates the results when a magnetomotive force of 100 A is applied to the core. As expected from the previous discussion, the flux density is confirmed to be constant in all pieces of the core in the proposed structure regardless of the radial distance from the inner side, whereas the flux tends to be dense at the inner side and sparse at the outer side in the conventional structure. The differences are obvious in the proposed core structure especially at the corners.

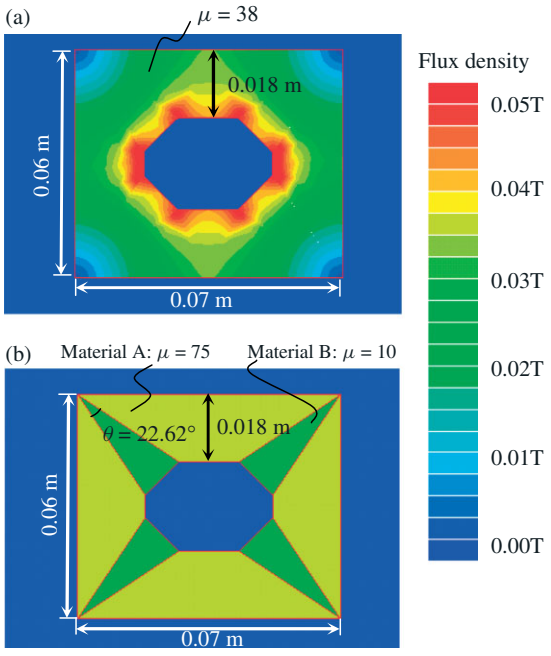


Fig. 4. Calculated flux distribution of conventional and proposed core structures. (a) Conventional core structure. (b) Proposed core structure

Although the proposed core has homogeneous density inside a piece of the core, the density is not necessarily the same between different pieces. This is because the width of the magnetic path expands at the corner. Equation (7) determines the ratio of the flux density between two neighboring pieces of the corner structure. The ratio is obtained for Fig. 4(b) by substituting $\theta_A = \theta/2 + 45^\circ = 56.31^\circ$ and $\theta_B = \theta/2 = 11.31^\circ$ into the equation:

$$\frac{|\mathbf{B}_B|}{|\mathbf{B}_A|} = \frac{\cos\theta_A}{\cos\theta_B} \approx 0.57 \tag{8}$$

The flux density of the material B is 0.57 times that of material A.

Since material A is more deeply saturated than material B, the saturation property of material A tends to dominate the whole inductor.

3.2. Saturation property The proposed structure is expected to have better saturation property than the conventional structure because the flux concentration at the inner side is solved and the excessive increase of magnetic resistance is suppressed there. This subsection employs numerical calculations to evaluate the effect by comparing the current and the flux at magnetic saturation between the inductors made of the proposed structure and the conventional structure.

In this paper, the saturation current and flux are defined as the current and flux when inductance decreases by 40% of the initial value. The inductance decrease by magnetic saturation is calculated using the simulator mentioned in the previous section.

The saturation property is dependent not only on flux distribution but also on the core dimensions, number of windings, the AL value, and the saturation flux density of each constituent core material. The undesired dependencies are eliminated by applying the same core dimensions, the same number of windings, and the same AL value to both cores. Furthermore, the saturation property of a core material is given so that all materials have the same saturation flux density.

Specifically, the horizontal core dimensions are set the same as in the previous subsection. The heights of the cores are set at 0.035 m. The number of windings is set at 30. And the same AL value is set by employing the permeability shown in Fig. 4 as the initial permeability of each constituent material.

To give the same saturation flux density to the materials, we assumed the material permeability μ to decrease according to the following equation:

$$\mu = \frac{dB}{dH} = \frac{\mu_R \mu_V}{1 + (B/\beta)^\alpha} \tag{9}$$

where B and H are the flux density and the magnetic field intensity, respectively; μ_R is the initial relative permeability of the material; μ_V is the permeability of the vacuum; and α and β are constants.

Equation (9) indicates that each material saturates to the same degree by induction of the same flux density. This feature represents the saturation property of materials with the same saturation flux density.

We chose $\alpha = 2.5$ and $\beta = 0.4$, so that (9) yields a saturation property similar to actual iron powder core. Figure 5 illustrates the saturation property given by (9) and that of the iron powder core No. 52 and No. 2 [9] of Micrometals Inc. Material A in Fig. 4(b) corresponds to the No. 52 material, and material B to the No. 2 material. Figure 5 shows that (9) yields a property similar to that of the actual material.

The results of the numerical calculation are shown in Fig. 6. The averaged flux density is the total flux divided by the cross-sectional area of the magnetic path. By comparing the horizontal axis when the inductance decreases by 40% of its initial value, both the flux density and DC current improve by 16%. Since the DC current is proportional to the flux density in the first order

19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

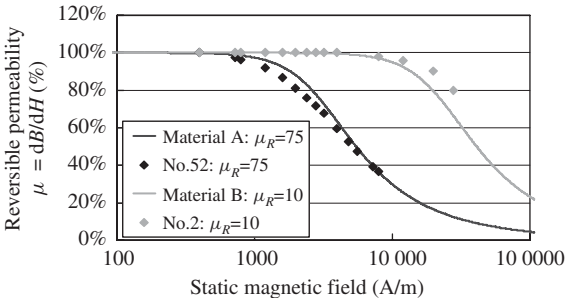


Fig. 5. Assumed saturation property of Materials A and B and Micrometals No. 52 and No. 2 (Modified from datasheet [9])

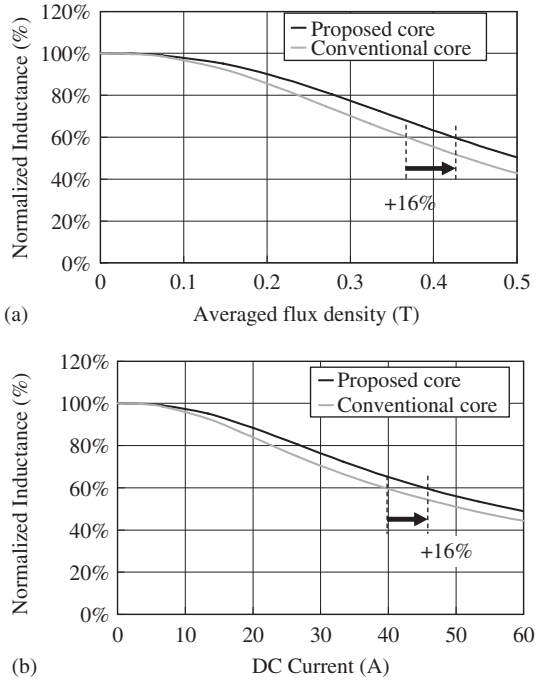


Fig. 6. Calculated saturation property of conventional and proposed core structures. (a) Normalized inductance versus average flux density. (b) Normalized inductance versus DC current

of approximation, it is natural that both results exhibit the same degree of improvement. Thus, the numerical calculation predicts improvement in the saturation property of approximately 16%.

4. Experimental Results

To confirm flux homogeneity and improvement in the saturation property in an actual core, the following experiments are performed on the cores shown in Fig. 7. Figure 7(a) is the proposed structure and Fig. 7(b) is the conventional structure. Both cores have the same outline and size as the numerical models shown in Fig. 4.

With a view to designing the cores to have similar composition, the iron powder core materials of Micrometals Inc. are employed for all core materials. Furthermore, material permeability is selected so that both cores have similar AL values.

4.1. Flux distribution In this experiment, flux distribution is compared at both the corner and straight magnetic path. The following method is employed to measure flux density distribution.

First, two lines of three small through-holes are made across the magnetic path of each test core in advance. As shown in Fig. 8, one line is made at the corner, and the other in a straight path. One-turn search coils are equipped so that each coil passes either

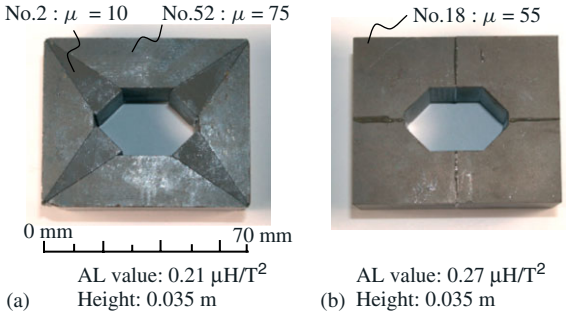


Fig. 7. Experimental cores. (a) Proposed structure. (b) Conventional structure

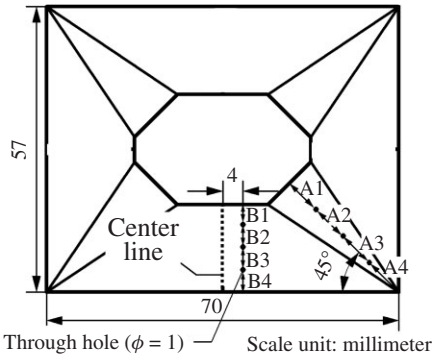


Fig. 8. Measuring points of flux distribution (A1–A4, B1–B4)

through a pair of neighboring holes or through the lateral vacant space and the nearest hole. Consequently, a series of search coils are obtained at the section A1–A4 and B1–B4 shown in Fig. 8.

Along with the search coils, a 10-turn magnetizing coil is equipped on the cores. The coil is wound so that the windings are distributed uniformly on the magnetic path. While the voltage shown in Fig. 9(a) is applied to the coil, the induced voltage Vr is measured for each search coil. As time integration of V_r yields the flux linkage ϕ of the search coil according to Faraday's Law, the flux waveform is obtained as shown in Fig. 9(b). Thus, the flux linkage $\phi_{\max}$ at point TA in time is obtained as follows:

$$\phi_{\max} = \max\left(\int V_r \mathrm{d}t\right) \tag{10}$$

Finally, the flux density at TA is obtained by dividing $\phi_{\max}$ by the core cross-sectional area S, which links to the search coil:

$$B = \frac{\phi_{\max}}{S} \tag{11}$$

To prevent the eddy current in the core material from disturbing the flux distribution, the voltage applied to the magnetizing coil is selected so that the magnetomotive force produced by the eddy current is much smaller than that by the coil. The order of the magnetomotive force of the eddy current is estimated based on the core loss characteristic. The details are shown in Appendix A2.

The results are shown in Fig. 10. The horizontal axis indicates a line of the search coils. The flux density of the proposed core structure is almost the same even in the outer and inner sides. This homogeneity is achieved both at the corner and the straight path, whereas the conventional core structure shows intense inhomogeneity at the corner.

The flux density at the corner is about half of that of the straight path in the proposed core. This is also consistent with the theoretical anticipation that the flux density at the corner is 0.57 times as large as that of the straight path, as mentioned in Section 3.1.

19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

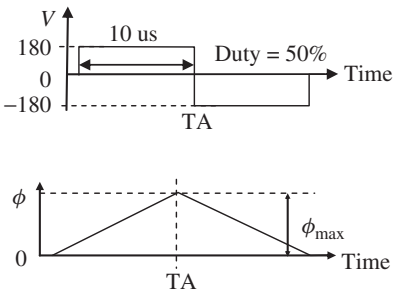


Fig. 9. Schematic diagrams of waveforms for the measurement of flux distribution

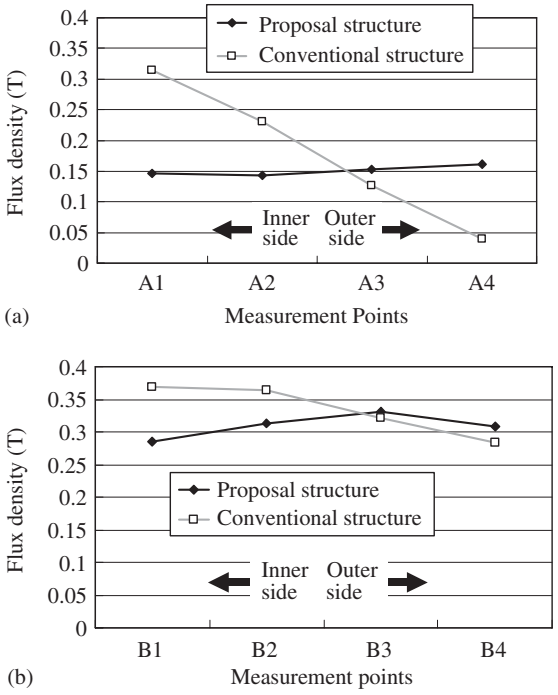


Fig. 10. Comparison of flux distribution between the inner and outer sides of magnetic path. (a) Corner path. (b) Straight path

4.2. Saturation property This section investigates the effect of homogenized flux distribution on the saturation property of the experimental cores. However, the saturation property is also dependent on its AL value and the saturation property of the core materials. Therefore, these effects should be eliminated either by using an evaluation method that is not sensitive to the undesirable dependencies or by canceling the effects using estimation. The former approach is employed for the AL value dependency, while the latter is used for the material dependency.

In general, the saturation property of an inductor is given as the inductance versus DC current characteristic. In this definition, an inductor whose core has a higher AL value shows a worse property because the same amount of magnetomotive force induces greater flux. This dependency can be eliminated by comparing the inductance decrease when the same amount of flux is induced in the core. Hence, the inductance versus DC flux characteristic is evaluated.

The following method is employed to measure the inductance and the DC flux. First, a 30-turn magnetizing coil is equipped on each experimental core. The coil is wound so that the windings are distributed uniformly on the magnetic path. Then, the voltage waveform shown in Fig. 11(a) is applied to the coil. The applied voltage is selected so that the eddy current does not affect the flux distribution, as discussed in the Appendix A2.

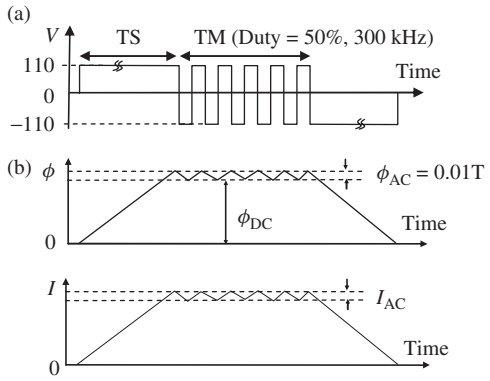


Fig. 11. Schematic diagrams of waveforms for the measurement of saturation property. (a) Voltage applied to an experimental inductor. (b) Flux wave form. (c) Current waveform

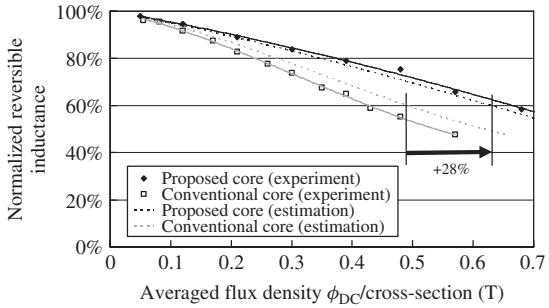


Fig. 12. Measured saturation property of the experimental cores. The dotted lines are the estimation when the same material property as No. 52 is applied to all materials constituting experimental cores

The total flux ϕ induced in the core is expressed as follows after Faraday's Law:

$$\phi = N^{-1} \int (V - RI)\mathrm{d}t \tag{12}$$

where N is the winding number, V is the voltage applied to the coil, R is its resistance, and I is the current.

Thus, the flux waveform is obtained as shown in Fig. 11(b). The DC flux is induced in the period TS, and then a triangular wave of small amplitude is overlapped in the period TM. The frequency of the overlapped wave is selected so that the flux amplitude ϕ_{AC} equals 0.01 T.

Finally, the inductance L is obtained from the flux amplitude ϕ_{AC} and the current amplitude I_{AC} after the following equation [10]:

$$L = \frac{N\phi_{\mathrm{AC}}}{I_{\mathrm{AC}}} \tag{13}$$

The measured saturation property of the experimental cores is illustrated by the solid lines in Fig. 12. The proposed core shows better saturation property than the conventional core. However, this improvement is partly due to the different saturation properties of the materials used in the experimental core. Thus, the following estimation is employed to cancel the effect.

First, the saturation properties of the core materials used in the experiment are investigated. This is experimentally obtained by measuring the saturation property of toroidal cores each made of a constituent material, according to the above-mentioned method. The results are illustrated in Fig. 13.

Among the three materials that constitute the proposed and the conventional core, the permeability decreases least in the No. 2 material. This difference, however, does not greatly affect the

19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

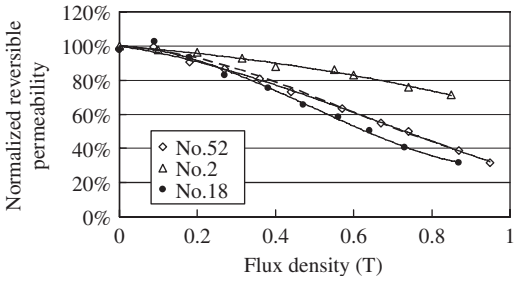


Fig. 13. Saturation property of core materials The dashed line is the adjusted property of No. 18 used in the estimation

saturation property of the proposed core because the flux density of the No. 2 material is about half of that of the No. 52 material and the No. 2 material hardly saturates in the proposed core anyway.

According to a simple estimation based on the permeance method, the inductance of the proposed core decreases by less than 5% if the saturation property of the No. 2 material is assumed to be the same as that of the No. 52 material. Details of the estimation are presented in Appendix A3. The estimated saturation property is represented by the black dotted line in Fig. 12. The estimated saturation property is obtained by multiplying the experimentally measured inductance by the inductance decrease rate L'/L obtained by (A25).

The conventional core also needs to adjust its material property to the No. 52 material. Since the flux lines do not have a simple shape in the conventional core, it is difficult to apply the permeance method. Hence, another method applicable to an arbitrary field distribution is employed instead.

Let $\mu_{18}(B)$ be the permeability of the No. 18 material expressed as a function of the flux density B. Then, we consider the conventional core whose material has been adjusted for permeability $\mu'_{18}(B)$ after the following equation:

$$\mu'_{18}(B) = \mu_{18}(B/1.15) \tag{14}$$

The property $\mu'_{18}(B)$ is almost the same as that of the No. 52 material, as illustrated by the dashed line in Fig. 13. Therefore, the property of the adjusted core under consideration is what we need to estimate.

As shown in Appendix A4, the adjusted core shows the same inductance as the original core when the flux density of the core is 1.15 times that of the original core. Hence, the saturation property of the adjusted core is estimated as illustrated by the grey dotted line in Fig. 12.

Finally, the improvement due to flux homogenization is evaluated by comparing the black and grey dotted lines in Fig. 12. As a result, a 28% improvement is found in the DC flux density when the inductance decreases by 40% of its initial value. Thus, the improvement is found not only by numerical calculation but also by experimentation.

This improvement is larger than that from the numerical results. The reason is not clear at present. However, one possible reason might lie in nonlinear material properties that were ignored in our theoretical and numerical investigation.

For example, some iron powder cores, including those used in our experiment, show larger permeability for large AC magnetic induction than for small AC induction [9]. Consequently, effective permeability is probably larger than the numerical model during the TS period in Fig. 11, where large amplitude of magnetic induction takes place. This enhancement of permeability occurs intensely at the inner flux paths and weakly at the outer flux paths of the conventional core because of the local inhomogeneity of magnetic induction. The difference, then, may promote the convergence of flux in the conventional core and help worsen its saturation property more than the numerical expectation, resulting in the large improvement found in the experimental results.

There also seems to be another nonlinear effect ignored in our investigation. As mentioned above, permeability of a magnetic material has nonlinearity due to magnetic saturation and AC flux induction. In cases where a large nonlinearity occurs, flux homogeneity is possibly not sufficiently obtained by the method presented in this paper because we assumed constant permeability to derive the turning angle of flux lines in (3). Therefore, adjustment may be required in calculating the angles θ_A and θ_B of the corner structure in the proposed core.

In future work, we will investigate the influence of nonlinearity on improvement of the saturation property of the proposed core and evaluate the effectiveness of adjusting the turning angles.

As for reduction of core materials, improvement in the saturation property enables the core volume to be downsized in order to achieve the same saturation property. This reduction can be roughly estimated by simple calculation.

As a case study, core reduction of an inductor made of the experimental core is calculated. We reduce the height of the proposed core without changing the initial inductance. Moreover, we preserve the total DC resistance of the magnetizing coil in order to not increase the copper loss. On the other hand, permeability of core materials is allowed to be adjusted without changing the saturation property of the materials. In other words, we assume the same permeability decrease rate under a given flux density as the original material. Then, we estimate the reduced height in order to achieve the same saturation current with the conventional core. Here, the saturation current is defined as the current when the inductance decreases by 40%.

According to calculations presented in Appendix A5, core reduction is estimated as 27% if we apply the experimental results of the improvement of the saturation property. If we apply the numerical results, there is an estimated 17% reduction of the core.

5. Conclusion

An iron powder core made of a single material and without gaps worsens its saturation property, because local magnetic saturation due to converging flux occurs at the inner side of the magnetic path. To avoid this problem, a novel core structure was presented that resolved the flux convergence and realized a homogeneous flux distribution. The proposed structure was composed of multiple materials, and polygonal flux lines were created by refraction at the material boundary.

The flux distribution of the proposed core structure was investigated using both numerical calculation and experimentation, and the theoretically expected distribution was successfully confirmed.

Furthermore, improvement of the saturation property was also investigated. As a result, improvement of approximately 16% was obtained from the numerical calculation and of 28% from the experimental results.

As for the core volume necessary to obtain the same saturation property, a 27% reduction is expected from the estimation based on the experimental results.

Appendix

A1. Shape of Flux Lines in Homogeneous Field

We consider a flux line in a homogeneous magnetic field, namely a field of constant flux density, inside a medium with constant permeability μ. If we define a 2-D Cartesian coordinate on a plane that includes the flux lines, the flux density vector is expressed as follows:

$$\mathbf{B} = (B_x, B_y) \tag{A1}$$

19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Let $k \neq 0$ be the norm of $\mathbf{B}$. Then

$$B_x^2 + B_y^2 = k^2 \tag{A2}$$

Differentiating the above equation partially with respect to x and y, we obtain the following equations:

$$B_x \left(\frac{\partial B_x}{\partial x}\right) = -B_y \left(\frac{\partial B_y}{\partial x}\right) \tag{A3}$$

$$B_x \left(\frac{\partial B_x}{\partial y}\right) = -B_y \left(\frac{\partial B_y}{\partial y}\right) \tag{A4}$$

Substituting (2) into (A3) and (A4), we have

$$B_x^2 \left(\frac{\partial B_x}{\partial x}\right) = B_y^2 \left(\frac{\partial B_y}{\partial y}\right) = (k^2 - B_x^2) \left(\frac{\partial B_y}{\partial y}\right)$$

$$\therefore \frac{\partial B_y}{\partial y} = \left(\frac{B_x^2}{k^2}\right) \left(\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y}\right) = 0 \tag{A5}$$

The rightmost equality in (A5) is derived using (1).

Because we can choose the coordinate system so that $B_x \neq 0$ and $B_y \neq 0$, the following equations are obtained by substituting (A5) into (A3) and (A4).

$$\frac{\partial B_x}{\partial x} = \frac{\partial B_x}{\partial y} = 0, \frac{\partial B_y}{\partial x} = \frac{\partial B_y}{\partial y} = 0 \tag{A6}$$

Equations (A6) indicate that the direction of the flux density vector is constant. In other words, flux lines are straight in the field where the flux density is constant.

A2. Estimation of Eddy Current

This section estimates the maximum possible loss by the eddy current in the core, and then derives the order of the eddy current in the worst case. To simplify the discussion, we approximate that the cross section of the core is circular and that flux distribution is homogeneous in this estimation. We represent the flux density of the model according to that at the straight path of the experimental core.

First, we estimate the maximum possible loss by the eddy current. According to the catalog [9] of the core manufacturer, the core loss per unit volume p is expressed by the following equation:

$$p = f \cdot a(B_{pk}) + df^2 B_{pk}^2 \tag{A7}$$

where f is the frequency, a is a function of B_{pk}, B_{pk} is half of the peak-to-peak value of AC flux density, and d is a constant.

The loss due to the eddy current is proportional to f^2 [11]. Thus, the loss contributes to $df^2 B_{pk}^2$.

As the worst case, we regard that the current produces the whole of $df^2 B_{pk}^2$. In addition, we apply d of the No. 52 material, which has the largest d value, to the whole core. Then, the maximum possible eddy current loss P_{eddy} is estimated as follows:

$$P_{\text{eddy}} = dMf^2 B_{pk}^2 \tag{A8}$$

where M is the volume of the core.

Next, we derive the relation between the total eddy current and P_{eddy}. Because we assume the constant value d inside the core, we are able to assume that conductivity is constant. We denote the conductivity by σ. According to electromagnetism [12], the eddy current density vector $\mathbf{i}_{\text{eddy}}$ is expressed by the following equation:

$$\frac{1}{\sigma}(\nabla \times \mathbf{i}_{\text{eddy}}) = \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{A9}$$

where $\mathbf{E}$ is the electric field vector, $\mathbf{B}$ is the flux density vector, and t is the time.

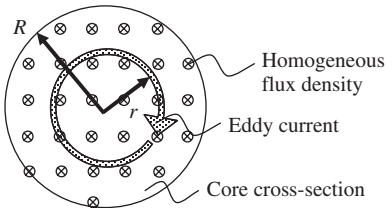


Fig. A1. Approximate model of the core cross section in the eddy current estimation

Now, we solve the above equation under the conditions of circular cross section and homogeneous flux distribution as shown in Fig. A1. Because of the axial symmetry of the problem, the eddy current forms a circular flow. Applying Stokes' theorem to the flow, we obtain the norm of the vector $\mathbf{i}_{\text{eddy}}$:

$$|\mathbf{i}_{\text{eddy}}| = \frac{\sigma}{2} r \frac{\partial \mathbf{B}}{\partial t} = \frac{\sigma}{2\pi R^2} \frac{V}{N} r \tag{A10}$$

where r is the distance from the center of the cross section, R is the radius of the cross section, V is the voltage applied to the magnetizing coil, and N is the winding number of the magnetizing coil.

The total eddy current I_{eddy} is estimated by introducing the length of the magnetic path λ:

$$I_{\text{eddy}} = \lambda \int_0^R |\mathbf{i}_{\text{eddy}}| \mathrm{d}r = \frac{\lambda \sigma}{4\pi} \frac{V}{N} \tag{A11}$$

Thus, we obtain the eddy current loss from (A11):

$$P_{\text{eddy}} = \lambda \int_0^R |\mathbf{i}_{\text{eddy}}|^2 \frac{2\pi r}{\sigma} \mathrm{d}r = \frac{\lambda \sigma}{8\pi} \left(\frac{V}{N}\right)^2 = \frac{I_{\text{eddy}}}{2} \frac{V}{N} \tag{A12}$$

Comparing this result with (A8), we obtain an estimation of the total eddy current.

$$I_{\text{eddy}} = 2 \frac{N}{V} dMf^2 B_{pk}^2 \tag{A13}$$

In the experiment presented in Section 4.1, $N = 10$, $V = 180$ V, $f = 5.0 \times 10^4$ Hz, $B_{pk} = 0.15$ T, and $M = 1.2 \times 10^{-4}$ m^3. The value d is given as 6.9×10^{-3} W/m^{3}Hz^{2}T^2 in the MKS system. Consequently, I_{eddy} is estimated as 5.2 A. Thus, I_{eddy} is ignorable because the magnetomotive force of the magnetizing coil is approximately 700 A.

As for the experiment in Sction 4.2, I_{eddy} is estimated as 1.1 A, since fB_{pk} is proportional to V/N and V/N is 0.2 times smaller than in Section 4.1. On the other hand, we have the same order of magnetomotive force of the magnetizing coil. Therefore, I_{eddy} hardly affects the flux distribution.

A3. Adjustment of the Saturation Property of the No. 2 Material

The estimation is based on calculating the inductance of the proposed core structure under given DC flux induction. First of all, we make an inductance model. As the first order of approximation, the change in permeability is assumed not to affect the flux distribution. Based on this assumption, the permeance method is employed. According to Ampere's Law, the integration of a magnetic field along a flux line C equals the magnetomotive force NI:

$$NI = \int_{C2} (B_2/\mu_2) \mathrm{d}l + \int_{C52} (B_{52}/\mu_{52}) \mathrm{d}l \tag{A14}$$

where μ_2 and μ_{52} are the permeability, $C2$ and $C52$ are the physical regions, and B_2 and B_{52} are the flux density of the No. 2 and No. 52 materials, respectively.

19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Note that B_2 and B_{52} are expressed by the total flux ϕ and core cross section S defined at the center line shown in Fig. 8:

$$B_2 = 0.57\phi/S, \quad B_{52} = \phi/S \tag{A15}$$

By dividing the whole core into flux paths with a tiny cross section dS and integrating (A14) over all paths, we obtain the following equation:

$$\begin{aligned}&\frac{1}{S\phi}\int\int_{C2}(B_2/\mu_2)\mathrm{d}l\,\mathrm{d}S + \frac{1}{S\phi}\int\int_{C52}(B_{52}/\mu_{52})\mathrm{d}l\,\mathrm{d}S\\ &= \frac{0.57}{S^2}\int\int_{C2}(1/\mu_2)\mathrm{d}l\,\mathrm{d}S + \frac{1}{S^2}\int\int_{C52}(1/\mu_{52})\mathrm{d}l\,\mathrm{d}S\\ &= \frac{0.57^2}{S^2}\frac{V_2}{\mu_2} + \frac{1}{S^2}\frac{V_{52}}{\mu_{52}} = \frac{NI}{\phi} = R\end{aligned} \tag{A16}$$

where V_2 and V_{52} are the volumes of the No. 2 and No. 52 materials, respectively, in the proposed core, and R is the total magnetic resistance. When we calculate the integral in (A16), we use the fact that μ_2 and μ_{52} are independent of the position inside the region of integration because of the uniform flux density.

By defining magnetic resistance R_2 and R_{52} contributed by materials No. 2 and No. 52 as (A17) below, (A16) can be reduced to (A18):

$$R_2 = \frac{0.57^2}{S^2}\frac{V_2}{\mu_2}, \quad R_{52} = \frac{1}{S^2}\frac{V_{52}}{\mu_{52}} \tag{A17}$$

$$R = R_2 + R_{52} \tag{A18}$$

Now, we introduce factors α_2 and α_{52}, which represent the permeability decrease rate by magnetic saturation, and express μ_2 and μ_{52} by their initial permeability $\mu_{2.0}$ and $\mu_{52.0}$, as follows:

$$\mu_2 = \alpha_2(B_2)\cdot\mu_{2.0}, \quad \mu_{52} = \alpha_{52}(B_{52})\cdot\mu_{52.0} \tag{A19}$$

Then, R_2 and R_{52} are expressed by $R_{2.0}$ and $R_{52.0}$, which are the magnetic resistance at sufficiently small DC flux induction, because the flux density is homogeneous in both materials.

$$R_2 = R_{2.0}/\alpha_2(0.57\phi/S), \quad R_{52} = R_{52.0}/\alpha_{52}(\phi/S) \tag{A20}$$

Because V_2, V_{52}, $\mu_{2.0}$, and $\mu_{52.0}$ are given for the experimental core, the ratio of $R_{2.0}$ to $R_{52.0}$ is obtained by calculating (A17):

$$R_{2.0} : R_{52.0} \approx 0.41 : 0.59 \tag{A21}$$

Thus, the magnetic resistance R under specific flux induction ϕ is obtained using (A20) and (A21):

$$R = \{0.41/\alpha_2(0.57\phi/S) + 0.59/\alpha_{52}(\phi/S)\}\cdot R_0 \tag{A22}$$

where R_0 is the total magnetic resistance at sufficiently small DC flux induction.

Since the inductance L is proportional to R^{-1}, the model of the proposed core is finally obtained:

$$L = \{0.41/\alpha_2(0.57\phi/S) + 0.59/\alpha_{52}(\phi/S)\}^{-1}\cdot L_0 \tag{A23}$$

where L_0 is the initial inductance.

To examine the model, we determine α_2 and α_{52} according to the properties of the No. 2 and No. 52 materials shown in Fig. 13, and estimate the saturation property of the experimental core using (A23). The estimation is then compared with the experimental results for the proposed core. The estimation is illustrated by the dotted line in Fig. A2. The comparison shows that the model explains the experimental result well in the first order of approximation.

Now, we adopt $\alpha_2 = \alpha_{52}$ and determine them by the property of the No. 52 material. This corresponds to the experimental core of the proposed structure when the No. 2 material has the same property as the No. 52 material. Then, its inductance L' is estimated by the following equation:

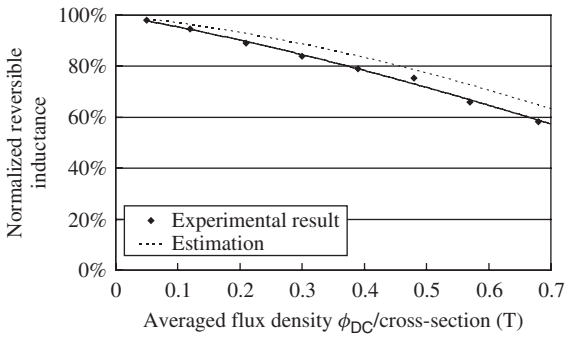


Fig. A2. Evaluation of the estimation model

$$L' = \{0.41/\alpha_{52}(0.57\phi/S) + 0.59/\alpha_{52}(\phi/S)\}^{-1}\cdot L_0 \tag{A24}$$

The effect of adjusting the saturation property of the No. 2 material is finally obtained:

$$L'/L = \frac{0.41/\alpha_2(0.57\phi/S) + 0.59/\alpha_{52}(\phi/S)}{0.41/\alpha_{52}(0.57\phi/S) + 0.59/\alpha_{52}(\phi/S)} \tag{A25}$$

The effect is smaller than a 5% decrease in inductance, since $0.95 < L'/L < 1$ when $\phi/S = B_{52} < 0.7$.

A4. Saturation Property of the Adjusted Conventional Core

Let $B(x)$ and $H(x)$ be the flux density and field intensity, respectively, realized in the experimental core of the conventional structure as functions of position x. Consider that the flux distribution $B'(x) = 1.15B(x)$ is induced in the adjusted core. Then, the field intensity $H'(x)$ is obtained as follows:

$$H'(x) = \frac{B'(x)}{\mu'_{18}(B'(x))} = 1.15\frac{B(x)}{\mu_{18}(B(x))} = 1.15H(x) \tag{A26}$$

Note that $B'(x)$ and $H'(x)$ are obtained by multiplying $B(x)$ and $H(x)$ by 1.15. If $B(x)$ and $H(x)$ are solutions of (1) and (2), then $B'(x)$ and $H'(x)$ are also a solution. Namely, $B'(x)$ and $H'(x)$ form an existing field.

Now, we denote the inductance, flux, magnetomotive force, and winding number of the experimental core by L, ϕ, F and N, respectively. Furthermore, we denote the corresponding value of the adjusted core with a prime symbol.

Since $B'(x) = 1.15B(x)$ and $H'(x) = 1.15H(x)$, we obtain $\phi' = 1.15\phi$ and $F' = 1.15F$. If $N = N'$, then

$$L' = N'^2\phi'/F' = N^2\phi/F = L \tag{A27}$$

This result shows that the adjusted core shows the same inductance as the experimental core when its flux is 1.15 times that in the experimental core.

A5. Estimation of Reduction Ratio of Core Volume

We denote the core height, inductance, flux, and winding number of the experimental conventional core as h, L, ϕ, and N, respectively. We denote the corresponding value of the proposed core with reduced height with a double prime symbol. First, we consider the requirement of the same saturation current. As a definition of inductance, we have the following relation [10]:

$$L = N\frac{\partial\phi}{\partial I}, \quad L'' = N''\frac{\partial\phi''}{\partial I} \tag{A28}$$

19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

where I is the current of the magnetizing coil.

By integrating the above equation with respect to I, we obtain the following equation:

$$N\phi_{\mathrm{sat}} = \int_0^{I_{\mathrm{sat}}} L\mathrm{d}I, \quad N''\phi''_{\mathrm{sat}} = \int_0^{I_{\mathrm{sat}}} L''\mathrm{d}I \tag{A29}$$

where I_{sat} is the saturation current, and ϕ_{sat} and ϕ''_{sat} are the saturation flux of the experimental conventional core and the proposed core with reduced height, respectively.

In this estimation, we approximate $L = L''$ for convenience, because we assume that $L = L''$ at $I = 0$ and $I = I_{\mathrm{sat}}$. Hence, we obtain the following relation:

$$N\phi_{\mathrm{sat}} = N''\phi''_{\mathrm{sat}} \tag{A30}$$

If we denote the saturation flux of the experimental proposed core as $\phi^{\circ}_{\mathrm{sat}}$, the saturation fluxes ϕ_{sat} and ϕ''_{sat} are expressed as follows:

$$\phi''_{sat} = \frac{h''}{h}\phi^{\circ}_{\mathrm{sat}} = \frac{h''}{h}\gamma\phi_{\mathrm{sat}} \tag{A31}$$

where γ is the improvement of the saturation property, given by the ratio $\phi^{\circ}_{\mathrm{sat}}/\phi_{\mathrm{sat}}$.

From (A30) and (A31), we obtain the following equation:

$$\frac{h''}{h} = \frac{N}{N''}\frac{1}{\gamma} \tag{A32}$$

Equation (A32) indicates that the reduction of the core volume is obtained if the ratio N/N'' is determined. We determine the ratio by the requirement of the same DC resistance of the coil.

For convenience, we assume a constant winding length per turn. We express the length per turn by adding a constant margin to the perimeter of the rectangular cross section of the magnetic path. Hence, the total length of the magnetizing coil is expressed as follows:

$$k = N(2w + 2h + m), \quad k'' = N''(2w + 2h'' + m) \tag{A33}$$

where k is the length of the wire of the coil, w is the width of the magnetic path, and m is the margin of the winding length.

Then, we consider the cross-sectional areas Σ and Σ'' of the wire of the coil. The area Σ is of the experimental conventional core, and the area Σ'' is of the proposed core with reduced height. Since the window area for the winding is the same for both cores, we assume the relation between Σ and Σ'' as follows:

$$N\Sigma = N''\Sigma'' \tag{A34}$$

Because the DC resistance is the same between the cores, we obtain the following relation:

$$\frac{k}{\Sigma} = \frac{k''}{\Sigma''} \tag{A35}$$

Substituting (A33) and (A34) into (A35), we obtain the ratio N/N'':

$$\frac{N''}{N} = \sqrt{\frac{2w + 2h + m}{2w + 2h'' + m}} \tag{A36}$$

Substituting (A36) into (A32), we obtain the following equation:

$$\gamma^2\left(\frac{h''}{h}\right)^2 = \frac{2w + m}{2w + 2h + m} + \frac{2h}{2w + 2h + m}\frac{h''}{h} \tag{A37}$$

$$\therefore \frac{h''}{h} = \frac{h + \sqrt{h^2 + \gamma^2(2w + m)(2w + 2h + m)}}{\gamma^2(2w + 2h + m)} \tag{A38}$$

To estimate the height reduction approximately, we give w and h according to values of the straight magnetic path. Then, $w = 0.018$ m, $h = 0.035$ m. And we take $m = 0.04$ m, for example.

According to the experimental results of saturation property improvement, we take $\gamma = 1.28$. We then obtain the reduction rate as $h'/h = 0.73$, which indicates that the proposed core structure reduces the core volume by 27%.

If we apply the numerical results, we take $\gamma = 1.16$. Then, we obtain the reduction rate as $h'/h = 0.83$, which indicates that the core volume decreases 17%.

References

(1) Yoshizawa Y, Oguma S, Yamauchi K. New Fe-based soft magnetic alloys composed of ultrafine grain structure. *Journal of Applied Physics* 1988; **64(10)**:6044-6046.
(2) Matsushita N, Kim D, Abe M. High-flux density and low-core losses attained for ferrite/permalloy composite core, *IEEE Transactions on Magnetics* 2006; **42(10)**:2824–2826.
(3) Long JG, McHenry M, Urciuoli DP, Keylin V, Huth J, Salem T E. Nanocrystalline material development for high-power inductors. *Journal of Applied Physics* 2008; **103(7)**:07E705-07E705–3.
(4) den Bossche AV, Valchev VC. *Inductors and Transformers for Power Electronics.* CRC Press: Florida; 2004.
(5) Matsuo Y, Kitaoka M, Nakao F. Present power supply transformers and their design. *Journal of the Magnetic Society of Japan* 2000; **24(4–2)**:831–834 (in Japanese).
(6) Landau LD, Lifshitz E M, and Pitaevskii L P. *Electrodynamics of Continuous Media* Butterworth-Heinemann: Oxford; 1984; 105-107.
(7) Kraus J D. *Electromagnetics.* McGraw-Hill: New York; 1991; 328–332.
(8) Born M and Wolf E. *Principles of Optics.* 6th ed. Cambridge University Press: Cambridge; 1980.
(9) Micrometals Inc. *Power Conversion and Line Filter Applications*, Issue L 2007; 26–28.
(10) den Bossche A V, Valchev V C. *Inductors and Transformers for Power Electronics.* CRC Press: Florida; 2004; 20-22.
(11) den Bossche A V, Valchev V C. *Inductors and Transformers for Power Electronics.* CRC Press: Florida; 2004; 112-117.
(12) Landau L D, Lifshitz E M, and Pitaevskii L P. *Electrodynamics of Continuous Media.* Butterworth-Heinemann: Oxford; 1984; 199-201.

Kazuhiro Umetani (Member) was born in Kobe, Japan, in 1980. He studied geophysical fluid dynamics at Kyoto University and obtained the PhD degree in 2007. He worked as a Circuit Designer with Toshiba for a year. From 2008, he has been engaged in power electronics research at Denso Corporation.



19314981, 2013, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/tee.21908 by Danish Technical Knowledge, Wiley Online Library on [13/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License